C^3: Committed Confidence Calibration Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

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Abstract

Abstract to be written.

1 Introduction

2 Setup

$$\mathrm{KL}\big(p^* \,\|\, \textstyle\prod_{i\in S} p_{\theta,i}\big) = \mathrm{TC}(x_S \mid x_{\text{obs}}) + \sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i}) \quad (1)$$

3 The instrument

4 Selection and confidence (C1)

5 Two sources of error (C2)

Committing a set S of masked tokens in one step samples them from the product of their marginals. Given the context x_{obs}, the KL from the true joint to that product splits exactly (Eq. 1, §2) into a dependency term (A) $= \mathrm{TC}(x_S \mid x_{\text{obs}}) = \sum_{i\in S} H(p_i^*) - H(p_S^*)$, which no product of marginals can represent, and an estimation term (B) $= \sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})$, the error of the marginals themselves (cf. Ben-Hamu et al., 2025). Work on parallel decoding targets (A) and takes the marginals as accurate (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026; Kang et al., 2026). We compute both exactly on a probe with known p^*, then test real-text proxies against it.

5.1 A JSON probe with known p^*

The probe fills the simple fields of 200 SchemaStore schemas (SchemaStore contributors, 2026), 6–16 per schema, with single-token values. Each field has a seeded Dirichlet(1) marginal, and fields are paired: with probability ρ, stated in a marker at the start of the instance, the follower is a fixed function of its leader's value, otherwise an independent draw, so p^* is known exactly. We fine-tune our 110M MDLM (§3) on 200k instances and evaluate 1,000 (200 per $\rho \in \{0, .25, .5, .75, 1\}$);

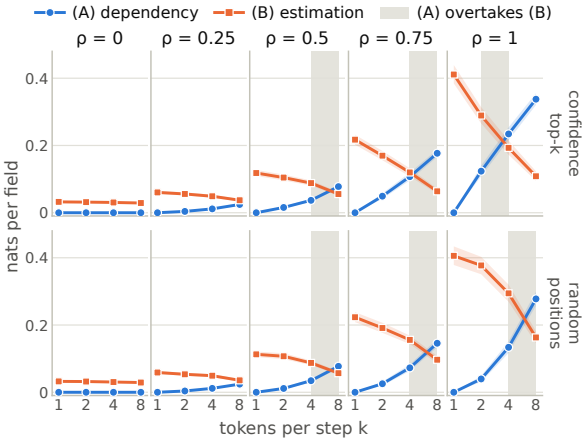


Figure 1: (A) and (B) per field on the probe, with 95% intervals, against tokens per step k (main model). Columns: ρ; rows: selection rule.

the main model is the fine-tune that never saw the marker masked (Appendix A.1). Decoding starts with every value masked. At each step the rule picks $k \in \{1, 2, 4, 8\}$ fields, the most confident (top-1 probability) or at random; we compute (A) and (B) for that set, with $p_{\theta,i}$ the model's probability of each allowed value (mass off the list counts against (B)), and fill it with the true values (teacher forcing). Terms are summed over steps, divided by the number of fields and bootstrapped over instances (10,000 resamples).

5.2 (A) overtakes (B) at high ρ and k

(A) is zero at $k = 1$ and at $\rho = 0$ and grows with both (Figure 1). For $\rho \geq 0.5$ it overtakes (B) at $k = 8$, or at $\rho = 1$ under confidence top-k already at $k = 4$; each crossover is also where the 95% interval of (A)−(B) clears zero. For $\rho \leq 0.25$ (B) dominates up to $k = 8$. At $\rho = 1$, $k = 8$ (confidence), (A) is 0.338 [0.320, 0.355] nats per field and (B) 0.109 [0.095, 0.123]: 76% of the total.

Confidence selection raises (A), not (B). The proposal expected ranking on noisy confidences

to inflate (B), but where (B) is higher than at random the intervals overlap. (A) rises instead ($\rho = 1$, $k = 2$: 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049]): the ranking commits leaders together with their followers more often than chance, which brings the crossover forward.

The terms trade off. ρ was meant to move only (A), but (B) rises too: at $k = 1$, from 0.032 at $\rho = 0$ to 0.411 at $\rho = 1$. The model reads the marker only in part (Appendix A.1), so a follower decoded after its leader misses the sharp marginal the copy implies; decoded with its leader, it has a broad true marginal, which the model matches, and the gap appears as (A) instead. At $\rho = 1$, from $k = 1$ to 8, (B) falls from 0.411 to 0.109 while (A) rises from 0 to 0.338. Without fine-tuning, (B) dominates everywhere ((A) is $\leq 4\%$ of the total).

5.3 Real text: proxies checked on the probe

These exact values hold for one synthetic distribution and one 110M model, not for real text, where p^* is unknown and we use proxies. For (A), parallel–sequential disagreement: the fraction of the k most confident masked positions whose token changes if they are committed one at a time (earlier ranks filled with the model's own tokens) instead of in one pass. For (B), the gap $\Delta = c - a$ (positive means overconfident) and the cross-entropy (CE) of the true token, at the same positions. We run the 110M model (fp32) and LLaDA-8B-Base (fp16; Nie et al., 2025) on 300 fit-half FineWeb-Edu documents (Penedo et al., 2024), one state at each mask rate $0.1, 0.2, \ldots, 1$ (Appendix A.1), and the same proxy on the probe.

The proxy tracks (A) weakly. Over the 15 confidence-rule cells with $k \geq 2$, its Spearman correlation with the exact (A) is 0.49 [0.28, 0.63] for the main model and 0.28 [0.06, 0.49] for the other. It rises with k, barely with ρ: at $k = 8$, 0.028 at $\rho = 0$, where (A) is zero, and 0.049 at $\rho = 1$.

On real text the (A) proxy is near zero. Over mask rates 0.1–0.9,[1] disagreement at $k = 8$ is 0.0021 for the 110M model and 0.0022 for LLaDA, and at most 0.0003 at $k \leq 4$. CE grows from 0.08 nats at $k = 1$ to 0.40 at $k = 8$ (LLaDA: 0.08 to 0.23), though it includes the text's own entropy and lower ranks are harder positions. Δ cannot be told from zero at $k = 1$ and is small but positive at $k = 8$: $+0.007$ [0.004, 0.010] and, for LLaDA, $+0.004$ [0.002, 0.007].

So at 2–8 tokens per step real text looks like the probe's low-ρ regime, where (B) dominates. Two limits: the proxy sees only dependence the model has learned, so near-zero disagreement does not show that (A) is small, and the proxies' units differ: real text gives a direction, not a split in nats.

[1] At rate 1 every input is the same all-`[MASK]` sequence. That one state gives the 110M model nearly all of its disagreement over all ten rates (0.0644 at $k = 8$); LLaDA does not disagree there.

6 C3

7 Related work

Limitations

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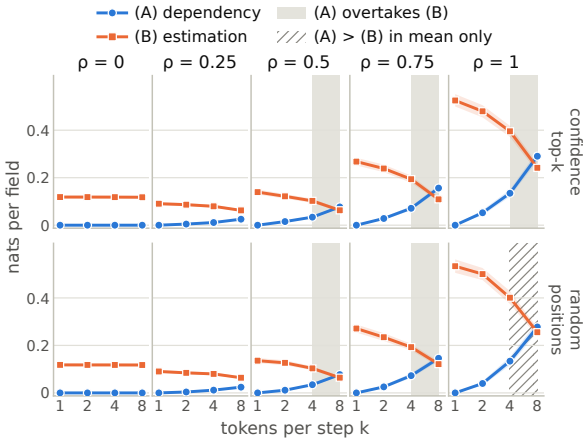


Figure 2: As Figure 1, for the fine-tune that saw the marker masked. Hatched: (A) $>$ (B) in mean only.

A Supplementary details

A.1 C2: the main model and T10 details

Main model. One fine-tune masked the ρ marker like any token (100M tokens); the main model never did, as at evaluation (300M). Neither moved the copied value's probability by our bar of 0.5 when the marker is switched from $\rho = 1$ to 0 with only the leader observed (0.014 and 0.251), so we took the lower validation NELBO (0.1751 against 0.1892 nats per token); the bar was first written down with these results, the fallback only when applied. The other (Figure 2) crosses over at $k = 8$ for every $\rho \geq 0.5$ under both rules.

T10. Unlike the plan, we skip documents under 256 tokens (fit-half ids up to 601; 600 for LLaDA) and keep at least 8 masked positions per state. Intervals resample documents (10,000 times).